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BITS Pilani, Hyderabad Campus
BITS F464 Machine Learning
Comprehensive Examination (Regular), First Semester (2023-24)

Date: Dec 6th, 2023

Time: 14.00 pm - 17.00 pm

Maximum Marks: 70

Instructions:

1. This is a closed book examination.
2. Answer all the questions.
3. Provide clear and concise answers. Answers should be limited to the space provided else will not be considered.
4. Make assumptions, wherever necessary and state the same clearly.
5. Use of calculators is allowed.

NOTE: To be updated only by the examiners

Q1 a (8 marks)	Q1 b (4 marks)	Q1 c (4 marks)	Q1 d (4 marks)
Q2 (15 marks)	Q3 (15 marks)	Q4 (20 marks)	Total (70 marks)

Q1.a) Consider the dataset below and estimate conditional probabilities to predict the class label for a test sample ($A=0$, $B=1$, $C=0$) using naive Bayes approach. [8 marks]

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

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Q1. b) List down the four steps carried out in Federated learning.

[4 marks]

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Q1. c) Mention any four recommended strategies to build real world ML systems. [4 marks].

Q1. d) Mention True or False for following statements. [4 marks]

- An ensemble tends to be more accurate than its base classifiers. The base classifiers may make mistakes, but the ensemble will misclassify X only if less than half of the base classifiers are in error.
- Ensembles yield better results when there is significant diversity among the models. That is, ideally, there is more correlation among classifiers.
- Each base classifier can be allocated to a different CPU and so ensemble methods are parallelizable.
- Random forests can be slower than either bagging or boosting because they consider fewer attributes for each split.

Q2. Consider the data given below, where each record represents weather condition and class label indicates whether people play or do not play in the given weather conditions.

Outlook	Temperature (Fahrenheit)	Humidity	Windy	Class
Sunny	75	70	TRUE	play
Sunny	80	90	TRUE	don't play
Sunny	85	85	FALSE	don't play
Sunny	72	95	FALSE	don't play
Sunny	69	70	FALSE	play
Overcast	72	90	TRUE	play
Overcast	83	78	FALSE	play
Overcast	64	65	TRUE	play
Overcast	81	75	FALSE	play
Rain	71	80	TRUE	don't play
Rain	65	70	TRUE	don't play
Rain	75	80	FALSE	play
Rain	68	80	FALSE	play
Rain	70	96	FALSE	play

a) Calculate the entropy and gini value of the given dataset.

[8 marks]

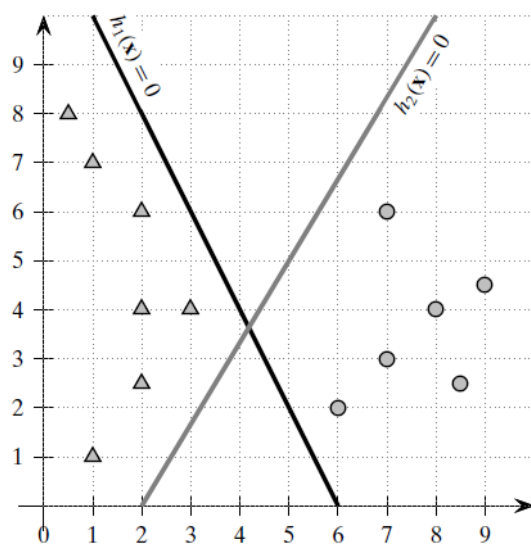
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b) Calculate the information gain for the split based on the “Outlook” attribute. [7 marks]

Q3. Consider the dataset in the figure below, which has points from two classes c_1 (triangles) and c_2 (circles). Answer the following questions.



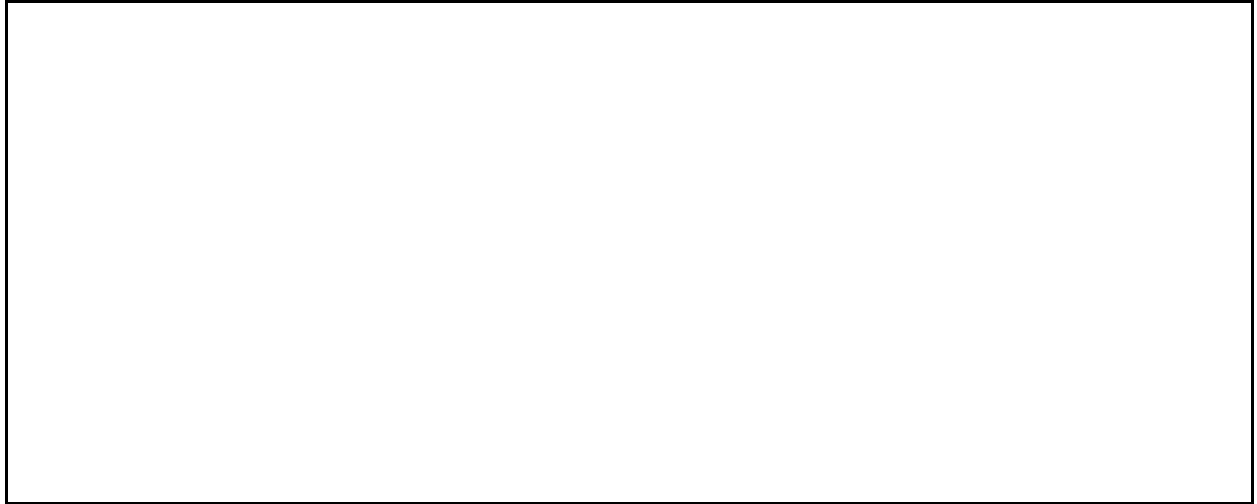
a) Find the equations for the two hyperplanes h_1 and h_2 .

[6 marks]

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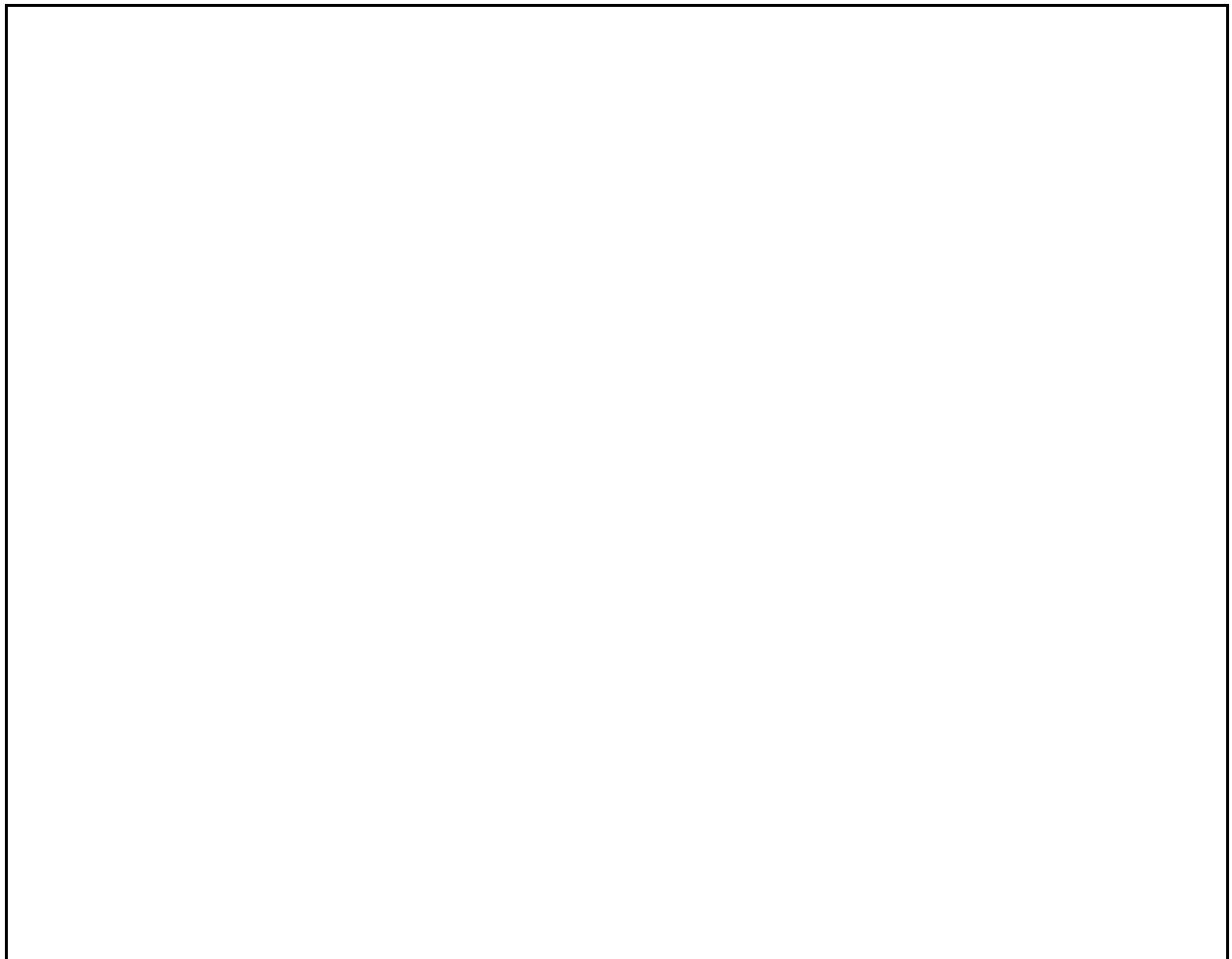
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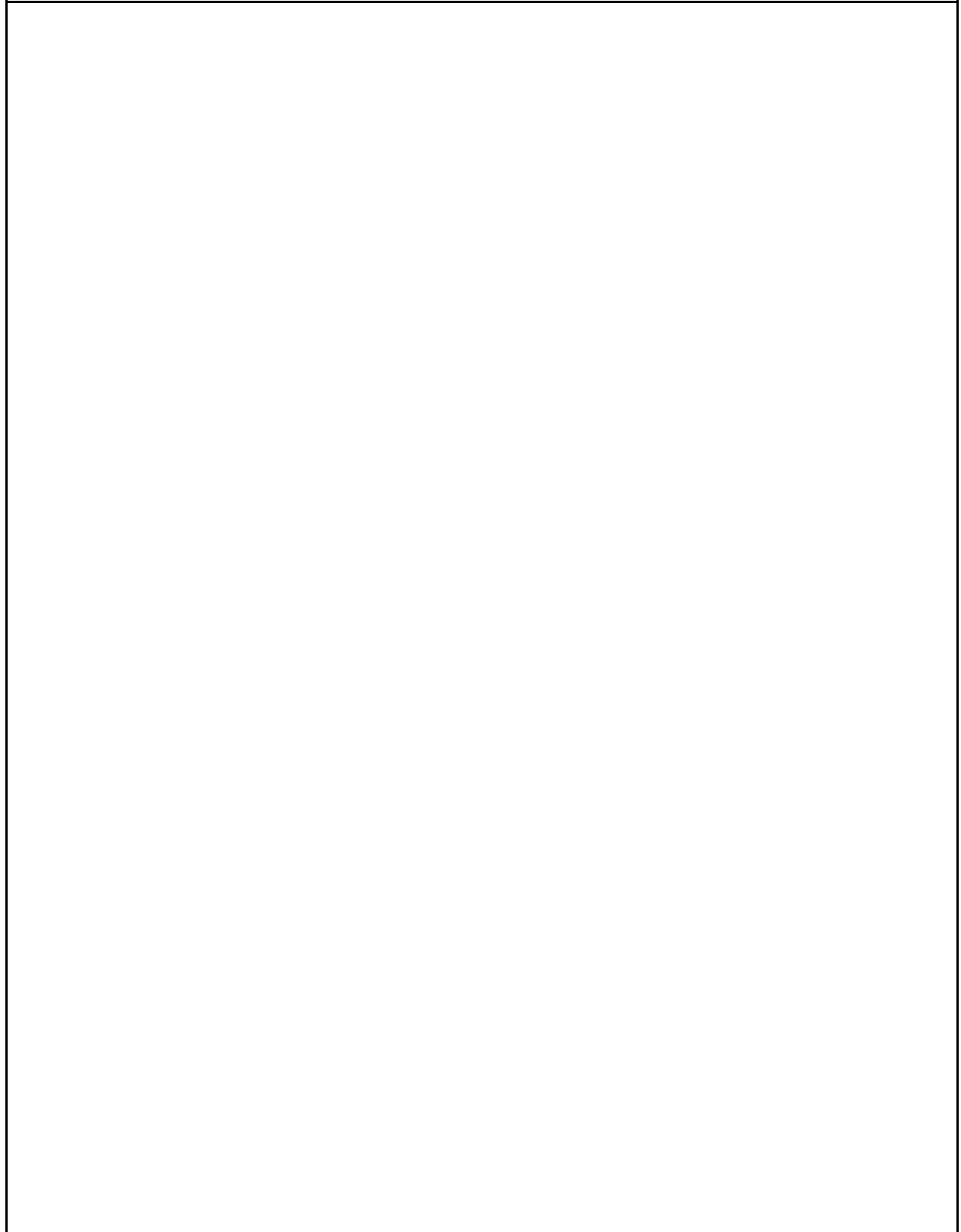
b) Show all the support vectors for h_1 and h_2 .

[5 marks]



c) Which of the two hyperplanes is better at separating the two classes based on the margin computation?

[4 marks]

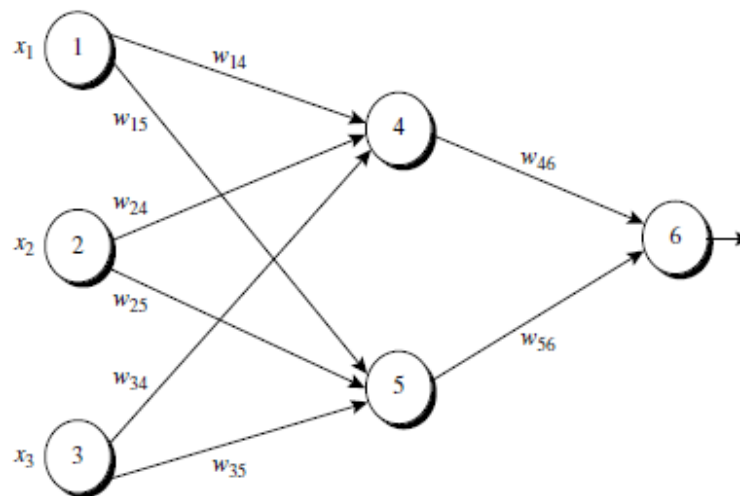


Q4. Consider a multilayer feed-forward neural network as given below.

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Initial Input, Weight, and Bias Values

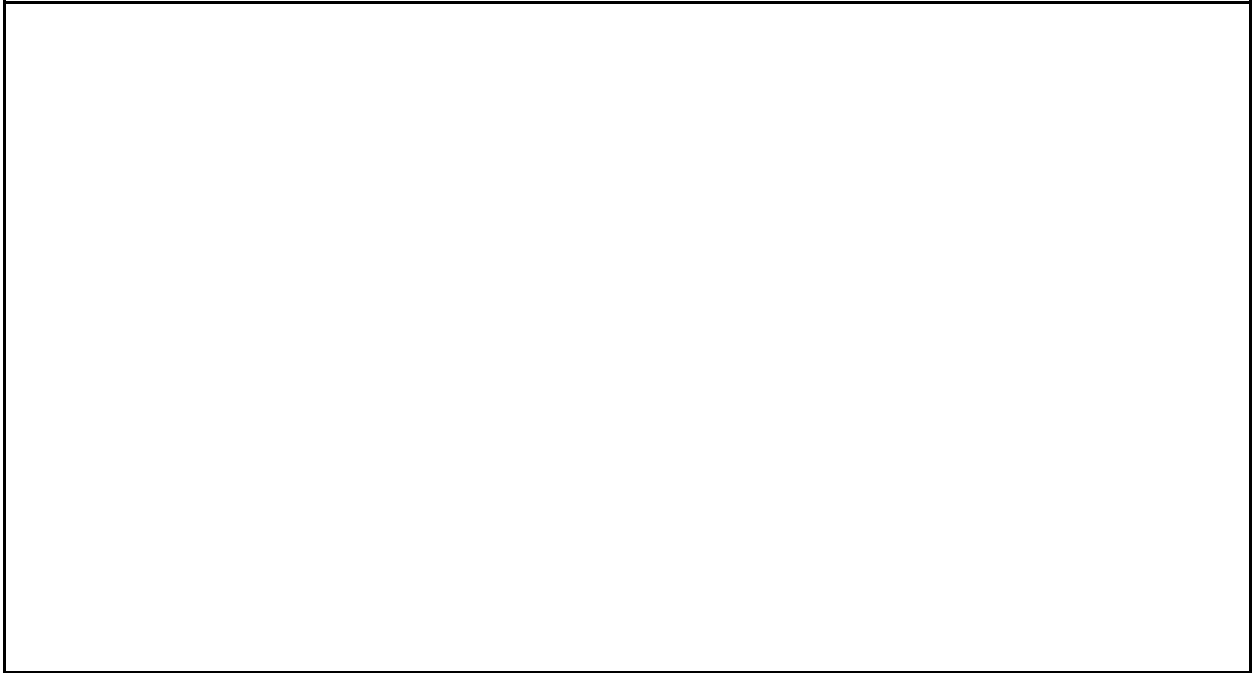
x_1	x_2	x_3	w_{14}	w_{15}	w_{24}	w_{25}	w_{34}	w_{35}	w_{46}	w_{56}	θ_4	θ_5	θ_6
1	0	1	0.2	-0.3	0.4	0.1	-0.5	0.2	-0.3	-0.2	-0.4	0.2	0.1

Let the learning rate be 0.9. The initial weight and bias values of the network are also given in the table above, along with the first training tuple, $X = (1, 0, 1)$, with a class label of 1 which is fed into the network. Assume logistic activation function for all nodes.

IMPORTANT NOTE: Please list the answers in a tabular format for all the subquestions below. Workings can be done in rough section.

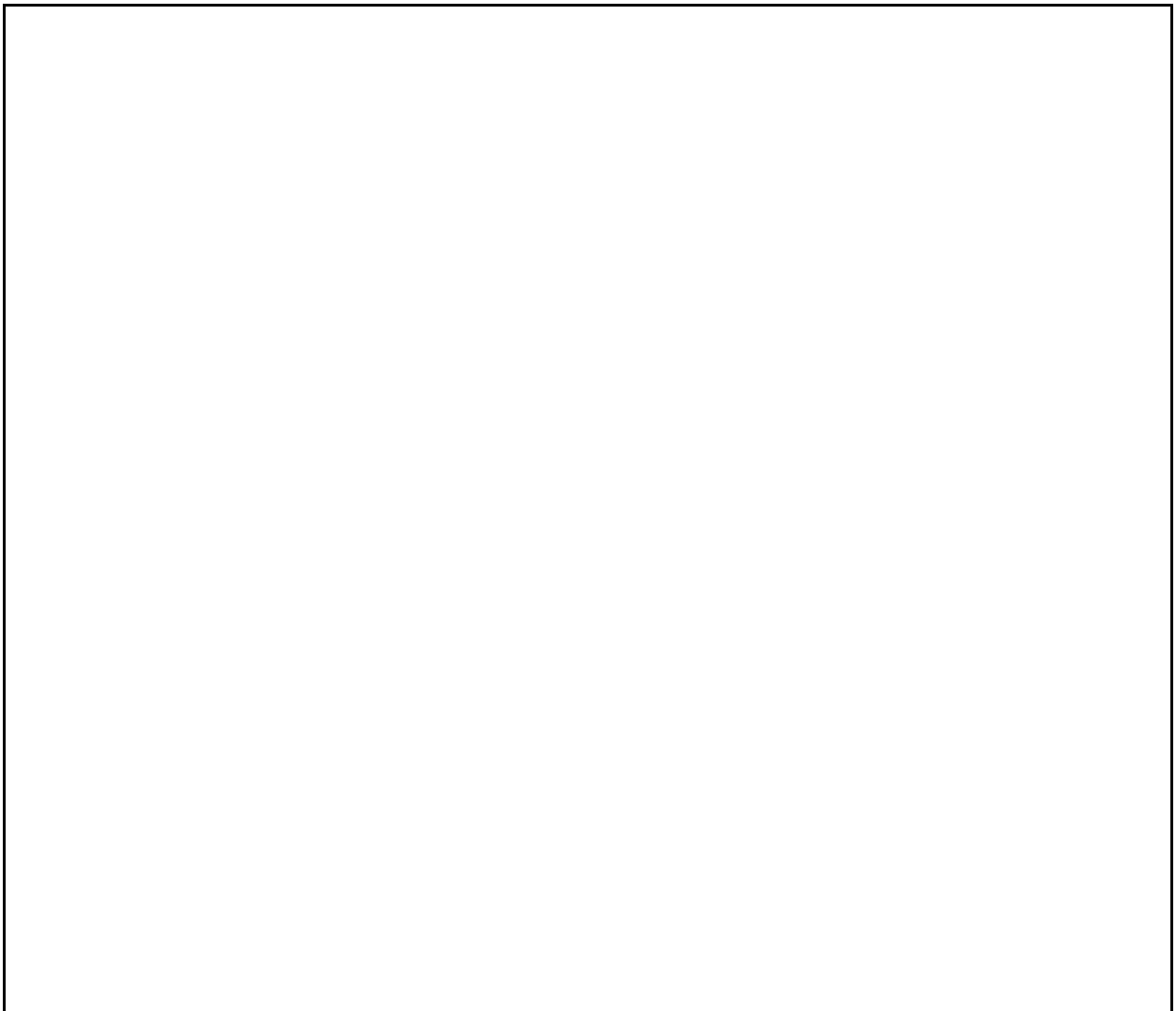
a) What is the net input and output of each node?

[6 marks]



b) What is the error of each node that is propagated backward?

[6 marks]



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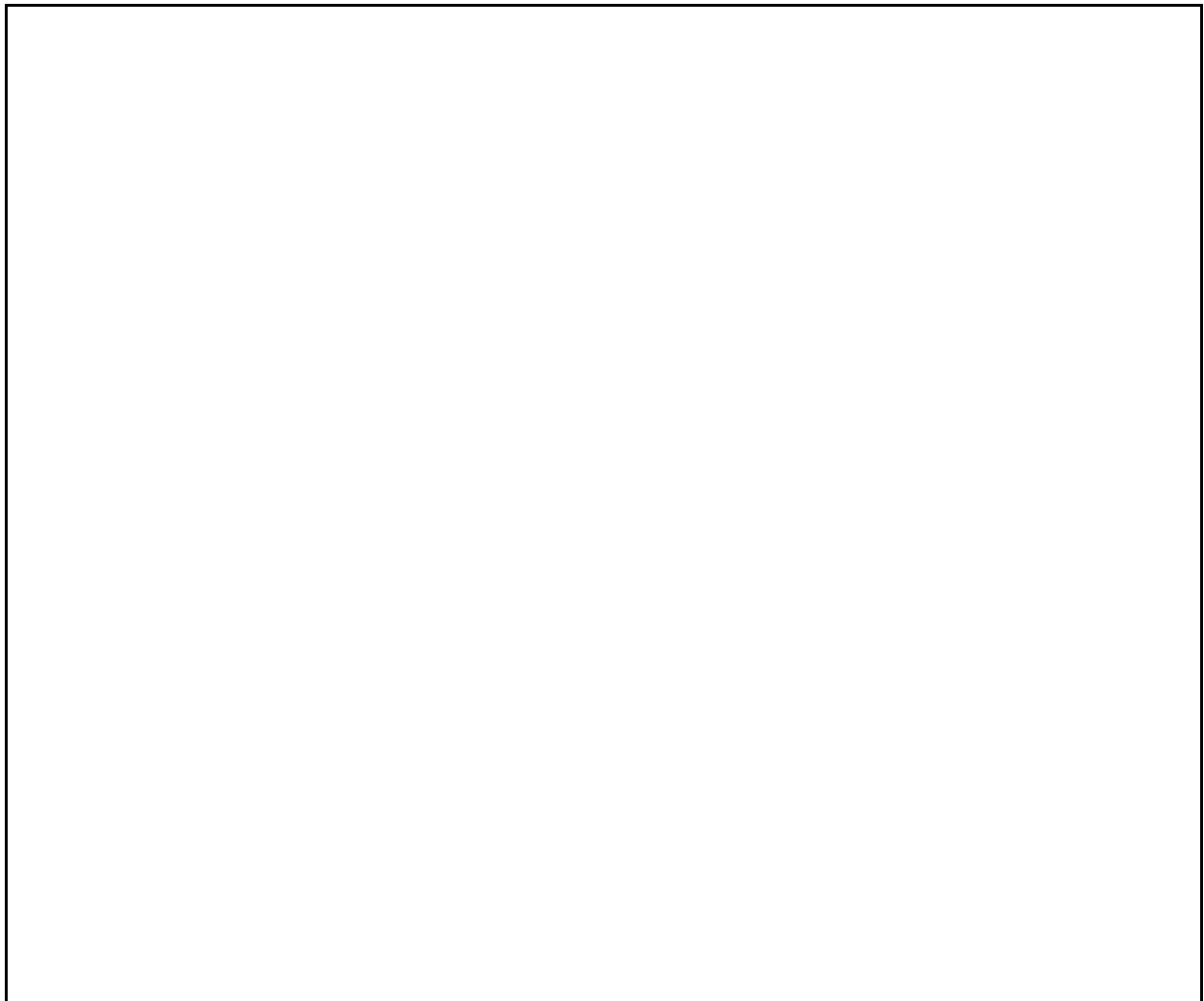
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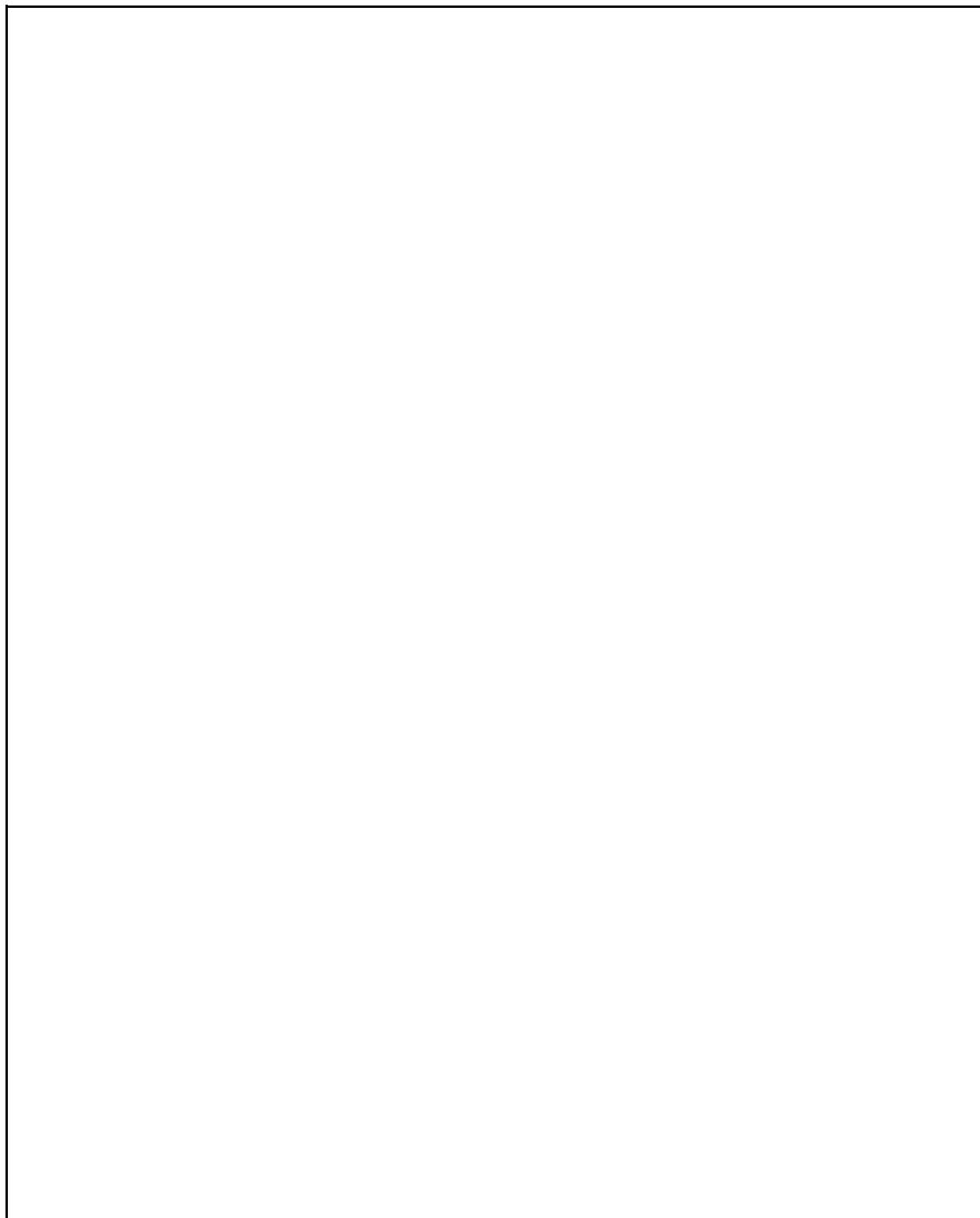
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c) What are the weight and bias updates?

[8 marks]

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